



Microbiology, Inflammatory Responses and Pathogenesis of Periodontal Diseases

Dr. Prathibha Nayak



Learning outcomes



- **Describe the microbiology of periodontium in health as well as gingival and periodontal diseases**
- **Describe the inflammatory responses in the periodontium along with the pathogenesis of gingival and periodontal diseases**

Introduction

- Within hours after birth, oral cavity is colonized by low numbers of mainly facultative and aerobic bacteria
- After exposure to *external environmental microbial sources and tooth eruption*, a more complex oral microbiota is established
- Bacteria live in *symbiosis adhering to hard or soft surfaces*
- Dental plaque contains more than *500 distinct microbial species*
- Bacteria may exist within the plaque biofilm or may be free floating (*planktonic*)

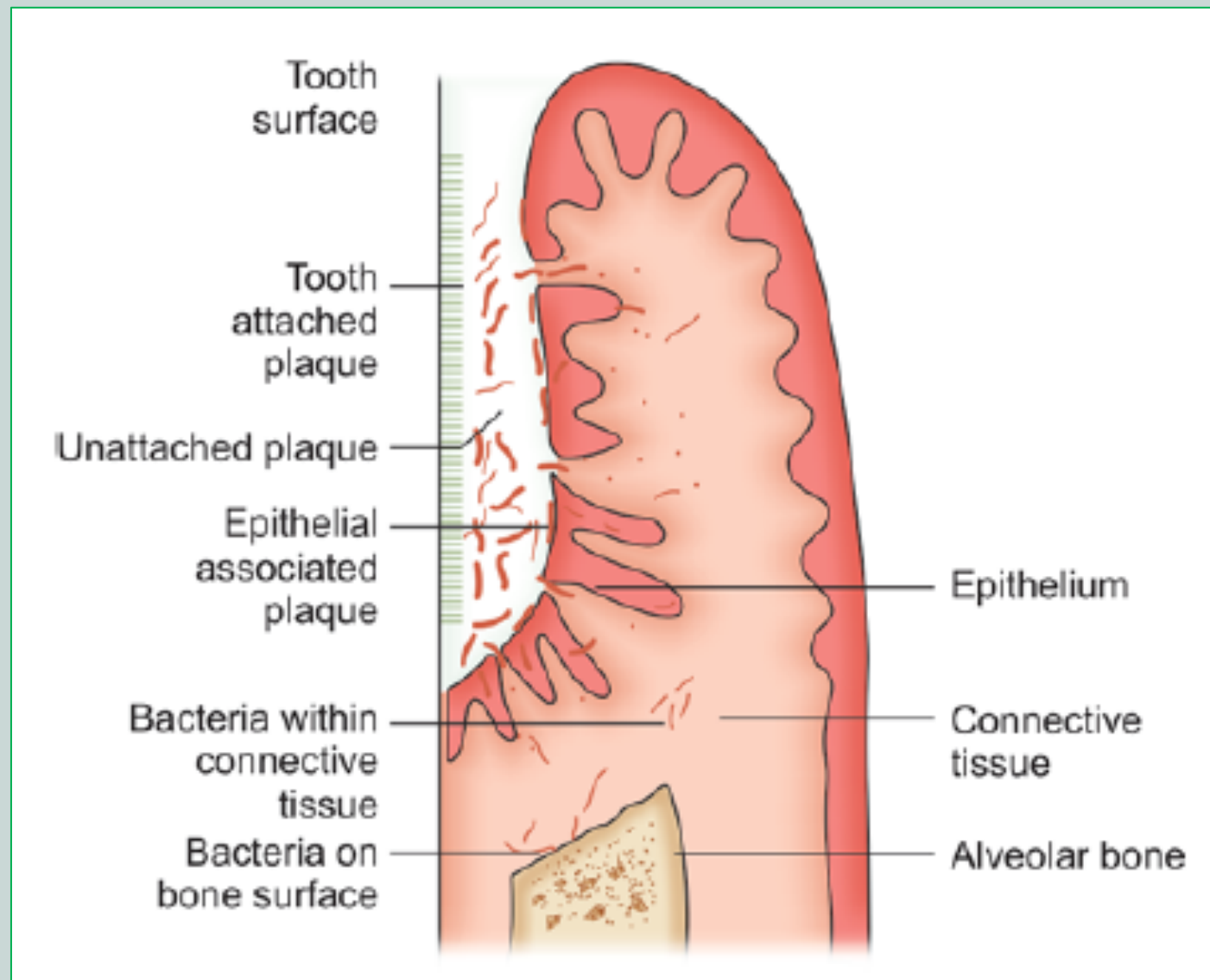
- ***Dental plaque*** is defined clinically as a structured, resilient, yellow-grayish substance that adheres tenaciously to the intraoral hard surfaces, including removable and fixed restorations
- Classified on the basis of its position from the gingival margin as:
 1. Supragingival plaque
 2. Subgingival plaque

- **Site specificity of plaque is significantly associated with diseases of the periodontium:**

1. *Marginal plaque-- initiation and development of gingivitis*

**2. *Supragingival plaque and tooth-associated subgingival plaque—
calculus formation and root caries***

**3. *Tissue-associated subgingival plaque--tissue destruction in
periodontitis***



Association of plaque and bacteria with tooth surface and periodontal Tissues

Microbiology of gingival and periodontal diseases

Periodontal Health

- ▷ Bacteria associated with periodontal health are primarily
 - a] *gram-positive facultative species*
 - b] *small proportions of gram-negative species*
 - c] *few spirochetes and motile rods*
- ▷ *Protective or beneficial species to the host*--- prevent the colonization or proliferation of pathogenic microorganisms
Eg: *Streptococcus sanguinis*

Transition From Health To Disease



- The ability of a bacterium to adhere to its host is important for induction of gingivitis or periodontitis
- Factors that determine whether active periodontal destruction will occur:
 1. *susceptible host*,
 2. *presence of pathogenic species*,
 3. *absence or small proportion of beneficial bacteria*
- Small amounts of plaque--controlled by body's defense mechanisms
- Dysbiosis-- periodontal destruction

Gingivitis



- **Subgingival microbiota consists of:**
 - equal proportions of gram-positive (56%) and gram-negative (44%)
 - equal proportions of facultative (59%) and anaerobic (41%) microorganisms
- Predominant gram +ve species--Streptococcus and Actinomyces
- *Pregnancy-associated gingivitis* -- Prevotella intermedia and Campylobacter rectus

Periodontitis



- Elevated proportions of *spirochetes*
- Anaerobic (90%) gram-negative (75%) bacterial species----
Porphyromonas gingivalis, *Tanerella forsythia*, *Prevotella intermedia*, *Fusobacterium nucleatum*, etc

Molar-incisor pattern periodontitis

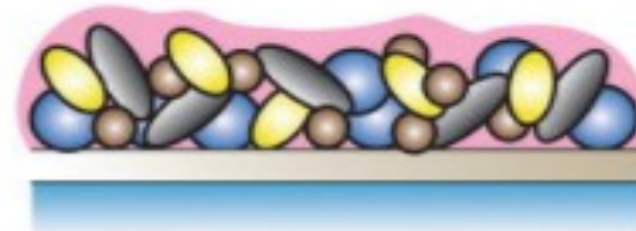


- *gram-negative, capnophilic (CO₂-requiring), anaerobic rods*
- *Aggregatibacter actinomycetemcomitans* comprise 90% of the total microbiota
- Other organisms include *P. gingivalis*, *F. nucleatum*, spirochetes etc.

Phase 1



Phase 2

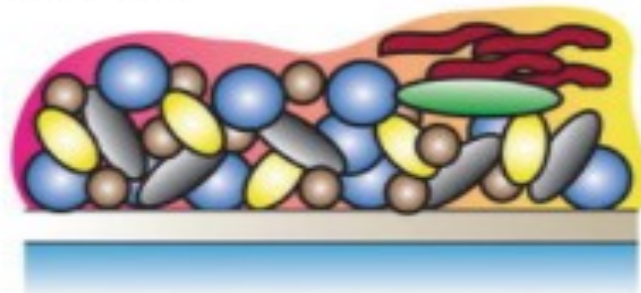


Mutualism and commensal bacterial growth



Gingivitis

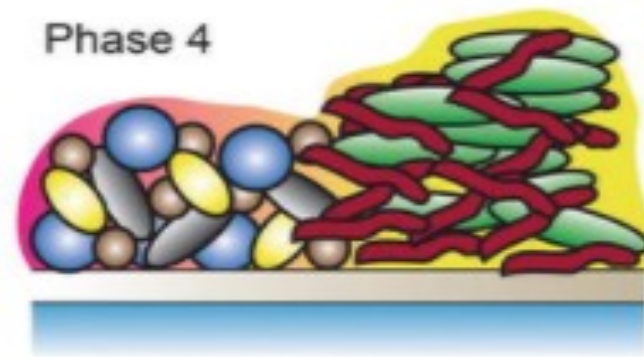
Phase 3



Transition bacterial growth

Periodontitis

Phase 4



Pathogenic bacterial growth



Necrotizing Periodontal Diseases

- Acute inflammation -- *necrosis of the marginal gingival tissue and the interdental papillae*
- Associated with *stress or HIV infection [immunosuppression]*, accompanied by *malodor and pain and possibly systemic symptoms*
- *Spirochetes and F. nucleatum* are present

Peri-implantitis



- An inflammatory process around an osseointegrated implant resulting in loss of supporting bone
- *Peri-implantitis* -- *A. actinomycetemcomitans*, *P. gingivalis*, *T. forsythia*, etc.

Inflammatory Responses And Pathogenesis Of Gingival And Periodontal Diseases

- Pathogenesis of a disease is the mechanism by which causative factor / factors causes the disease

Periodontal disease results from a complex interplay between the *subgingival biofilm and the host immune-inflammatory events* that develop in the gingival and periodontal tissues in response to the bacteria

Tissue damage---recognized clinically as *gingivitis /periodontitis*

Gingivitis precedes periodontitis, but not all gingivitis progress to periodontitis

- ▷ **Gingivitis---** inflammatory lesion is *confined to gingiva*
- ▷ **Periodontitis---** inflammatory processes extends to *periodontal ligament and the alveolar bone* resulting in attachment and bone loss
- ▷ The molecules that play a role in the pathogenesis of periodontitis:
 - 1. those derived from subgingival microbiota (microbial virulence factors)*
 - 2. those derived from the host immune–inflammatory response*
- ▷ Main determinant of susceptibility to disease is the *nature of the immune–inflammatory responses themselves*
- ▷ Defensive processes, which are *protective by intent* result in the majority of tissue damage that leads to the clinical manifestations of disease

Healthy Gingiva

- Gingival connective tissue contains some inflammatory cells, particularly *neutrophils*
- Neutrophils migrate → *junctional epithelium* → *sulcus/pocket*
- This low-grade inflammation occurs in response to the continued presence of bacteria and their products in the gingival crevice
- In addition, *lymphocytes and macrophages* also accumulate
- In clinically healthy tissues, this steady-state equilibrium between *low-grade inflammation* in the tissues and the continual presence of the bacterial biofilm may persist for many years or lifetime without clinical signs of gingivitis

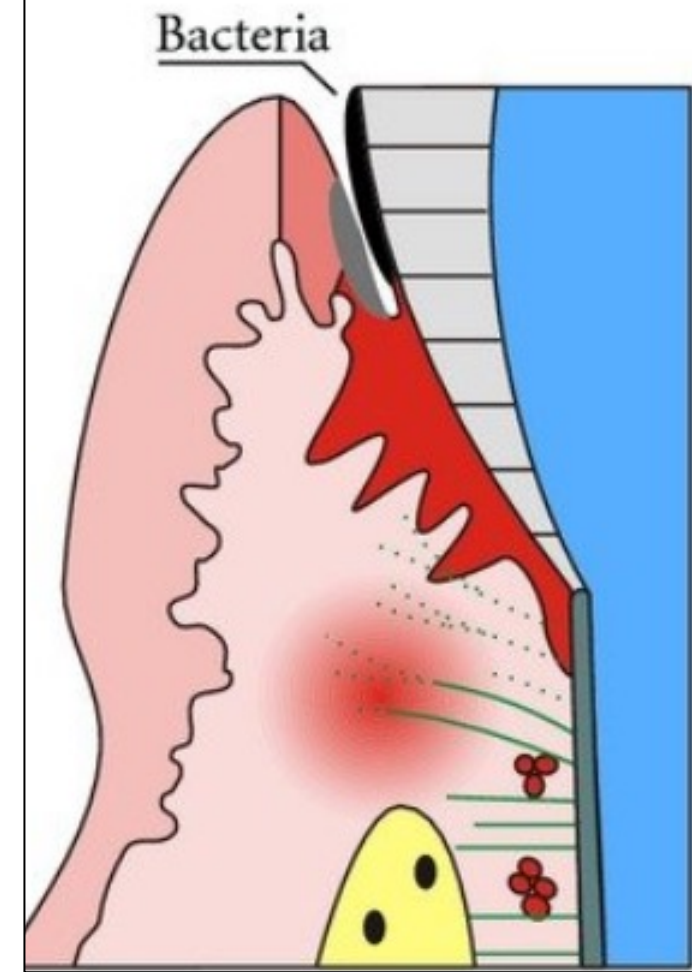
Gingivitis

- Plaque accumulation → inflammation and clinical signs of gingivitis
- Infiltration of the connective tissues by defense cells, neutrophils, macrophages, plasma cells, & lymphocytes

↓
Extracellular release of their destructive enzymes

↓
Collagen depletion and proliferation of the JE

- ***Vasodilation and increased vascular permeability*** -- leakage of fluid and passage of defense cells from the vasculature into the tissues-- enlargement of the tissues, which appear ***erythematous and edematous***

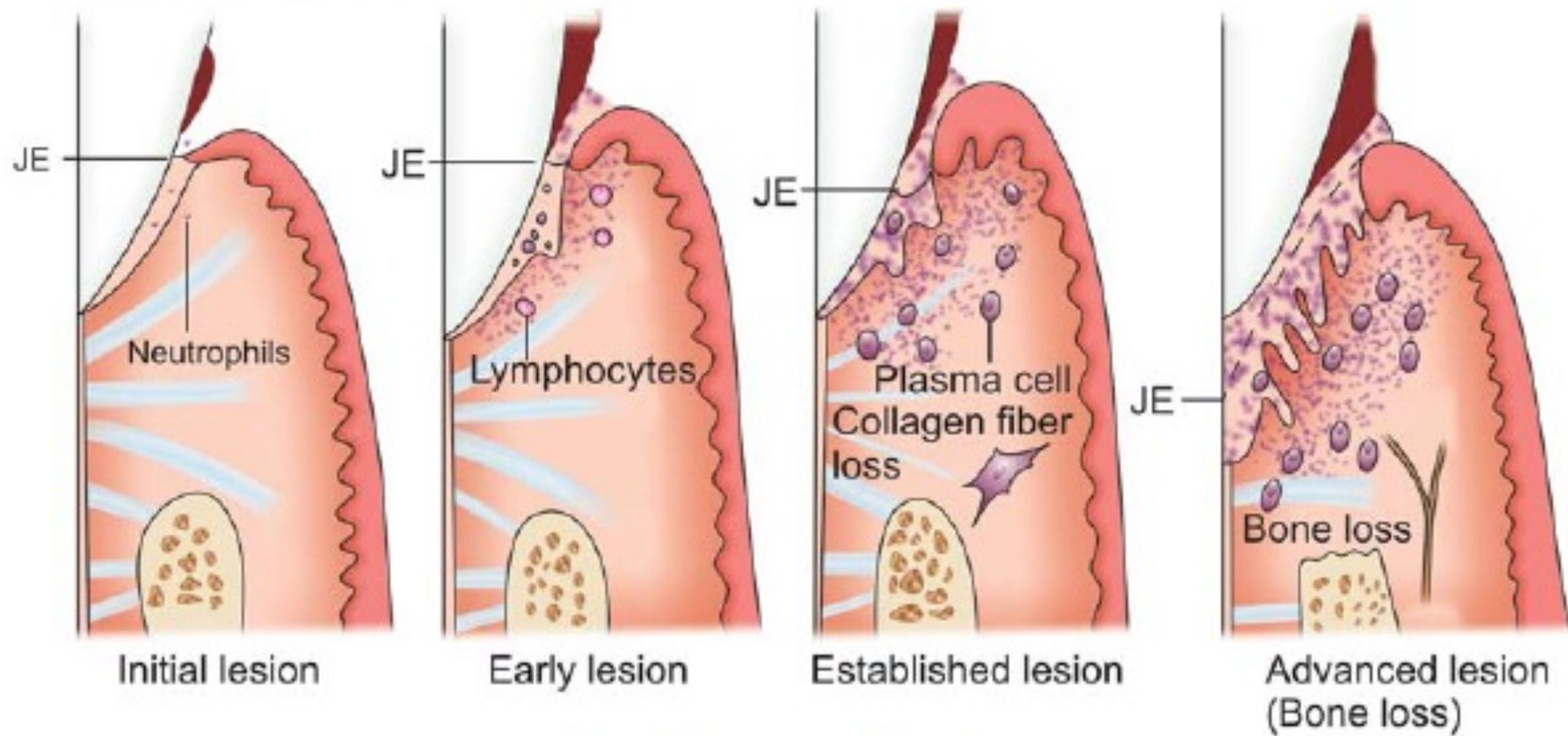


- ***Reversible*** if the bacterial challenge is reduced
- Histologic changes that occur in the gingival tissues are described as the ***initial, early, established, and advanced gingival lesions***
- Some individuals develop more pronounced gingival inflammation for a given plaque challenge and others develop minimal inflammation--- due to ***variations in host responses*** among individuals in terms of gingival inflammatory responses

Stages of gingivitis

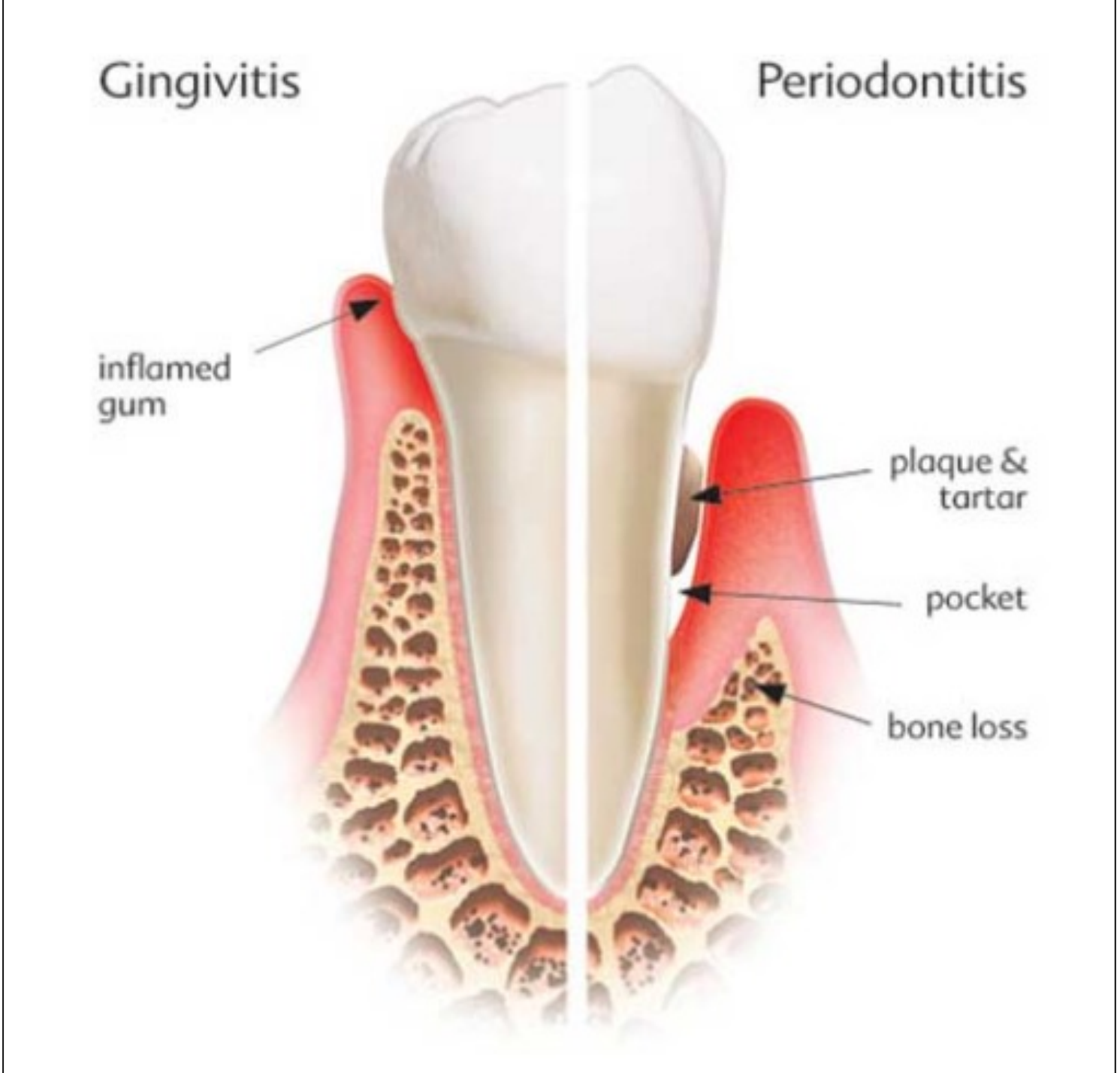
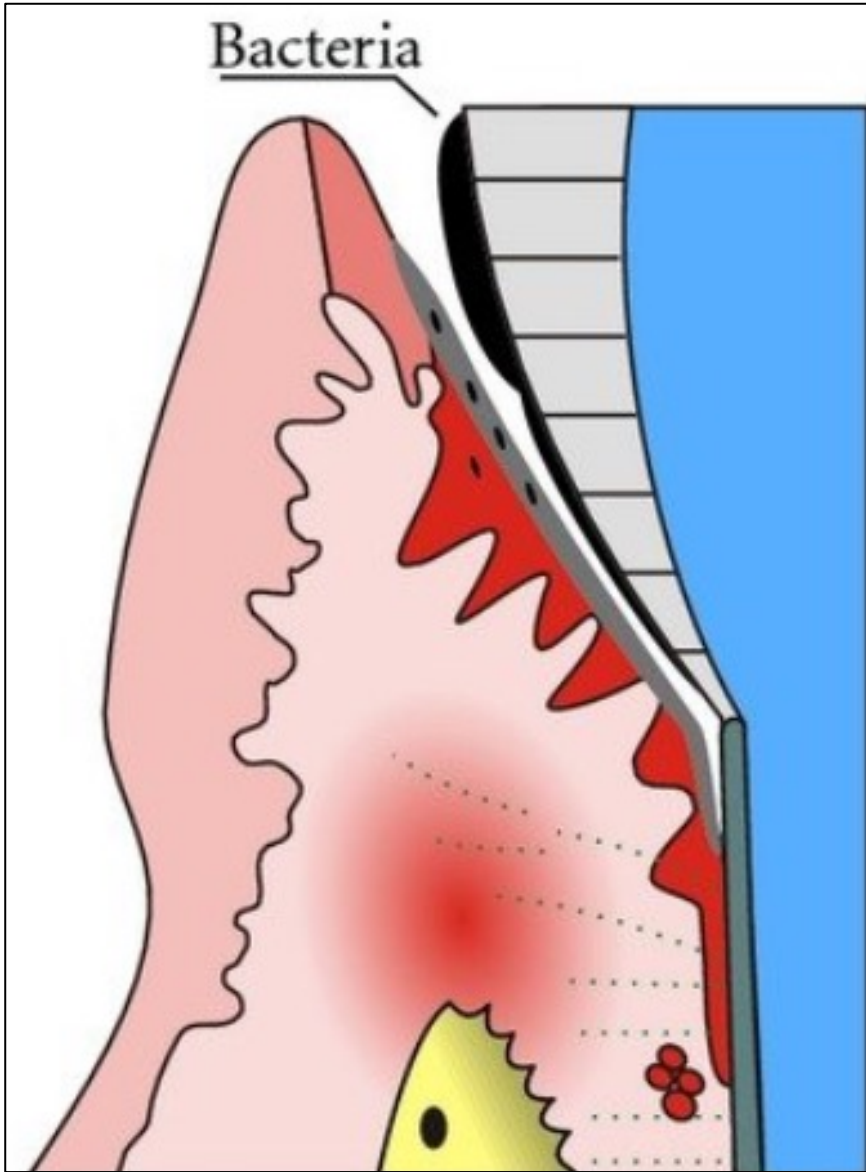
STAGE	TIME (DAYS)	PATHOLOGIC EVENTS	PREDOMINANT IMMUNE CELLS	CLINICAL SIGNS
Initial (Stage I) <i>Subclinical Gingivitis</i>	2-4	Vascular dilation Infiltration by PMNs into sulcular and junctional epithelium Perivascular loss of collagen	PMNs	Gingival fluid flow
Early (Stage II)	4-7	Vascular proliferation Infiltration by PMNs into sulcular and junctional epithelium T lymphocytes increase Retepeg formation Increased loss of collagen around the infiltrate	Lymphocytes	Erythema Bleeding on probing
Established (Stage III)	14-21	Same as stage II but more advanced Continued loss of collagen	Plasma cells	Erythema Bleeding on probing Changes in size, color, texture etc
Advanced (Stage IV)		Destructive changes extend into bone and other periodontal tissues	Plasma cells	Similar to stage III Attachment loss and bone loss

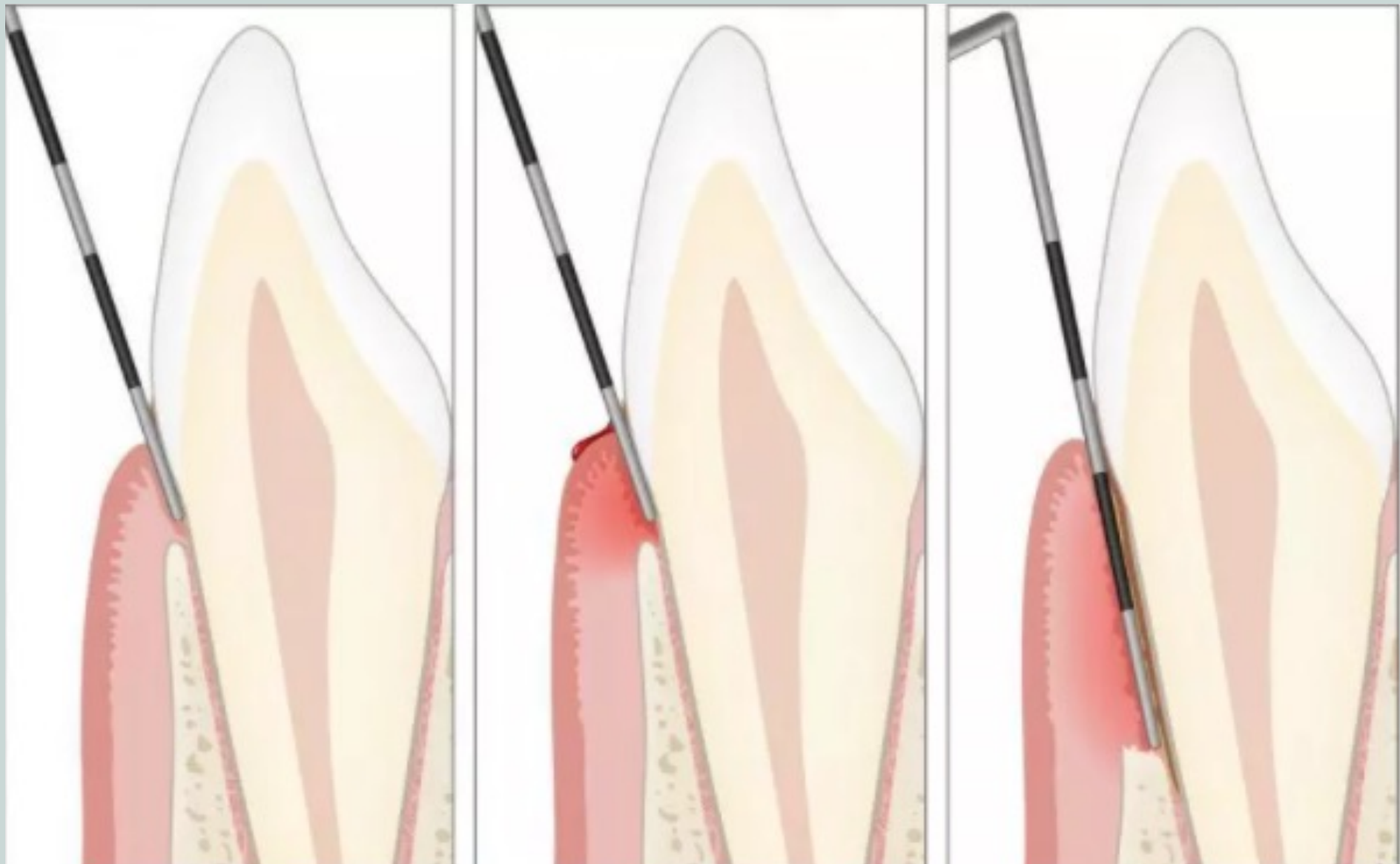
JE— Junctional epithelium



Periodontitis

- Determined by many factors-- *bacterial challenge, the host inflammatory response, and susceptibility factors (environmental and genetic factors)*
- Continued collagen destruction into the PDL and alveolar bone
- *Plasma cells* dominate in the connective tissues
- *Junctional epithelium migrates apically* along the root surface to maintain an intact epithelial barrier and form a *periodontal pocket*
- As pocket deepens, *plaque bacteria proliferate apically and are not* eliminated by the inflammatory response
- Bone retreats from the advancing inflammatory front as a defense mechanism to prevent the spread of bacteria into the bone----*bone loss*



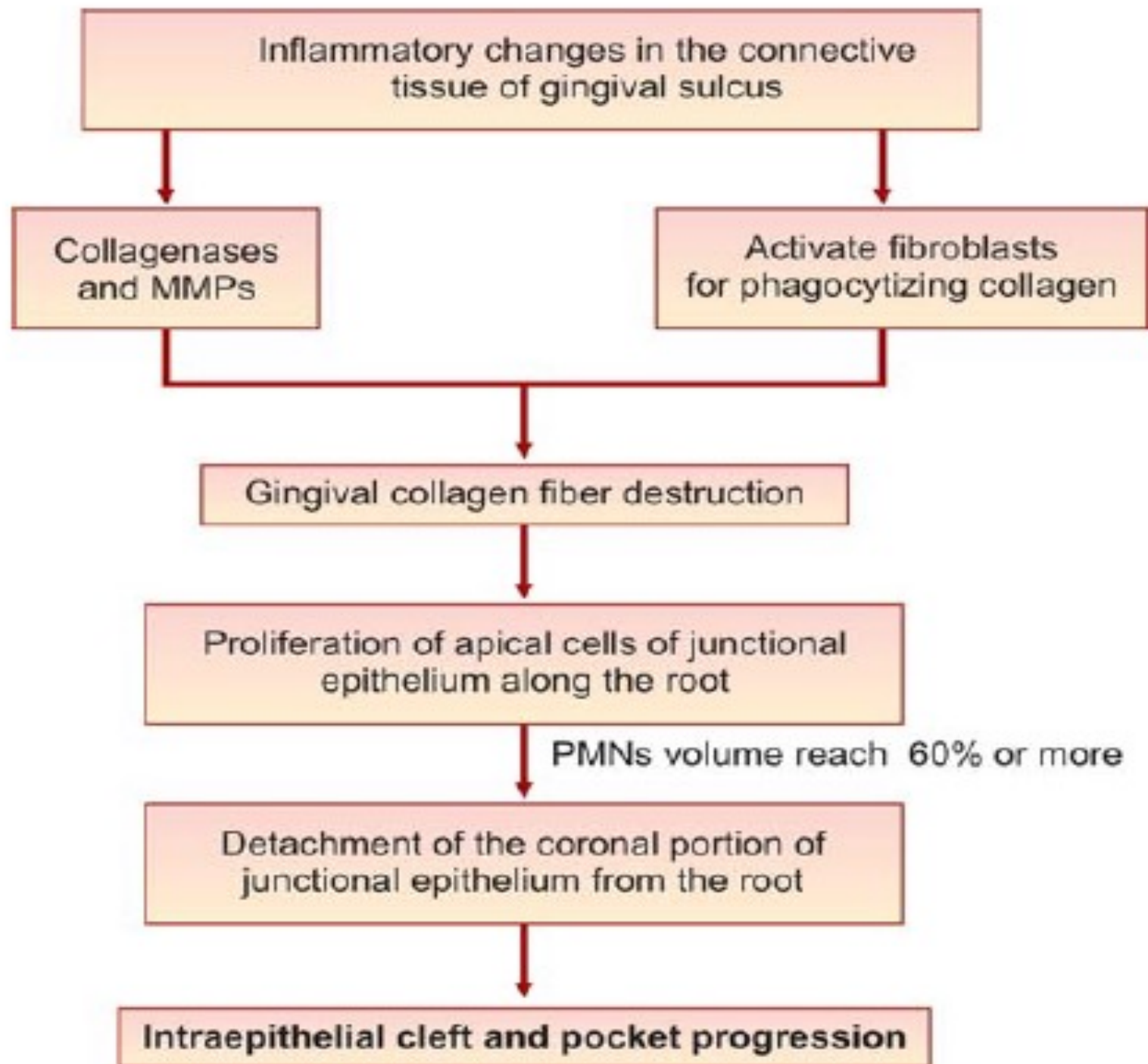


Healthy

Gingivitis

Periodontitis

How does a
periodontal
pocket develop?



Pathogenesis of Alveolar Bone Resorption

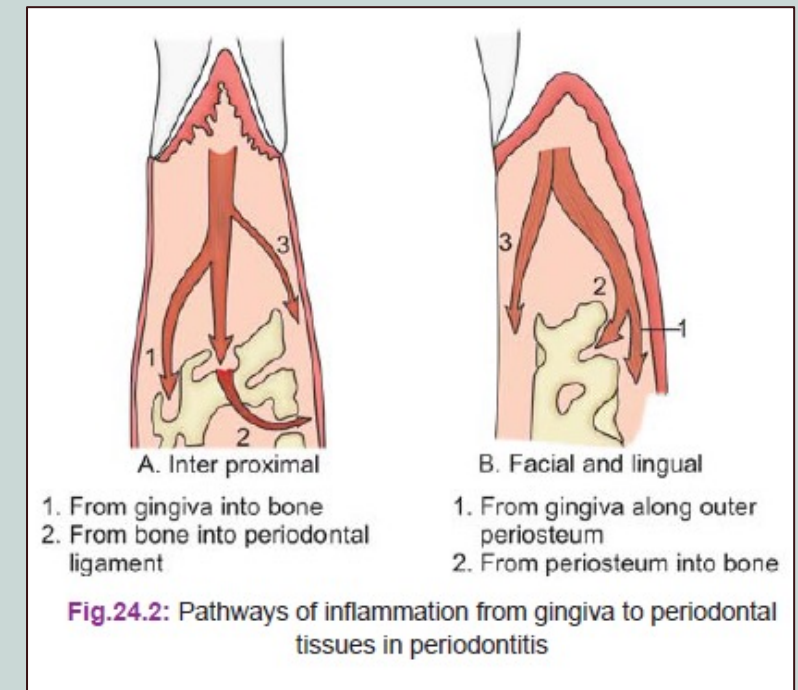
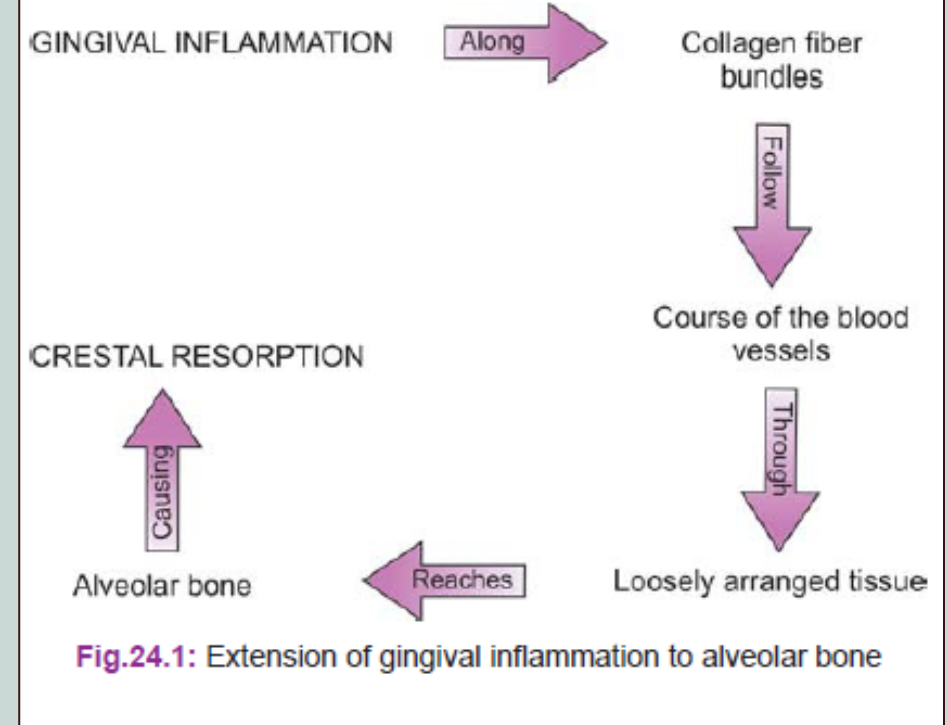
- Inflammation reaches the bone, it spreads into the marrow spaces



- *Increase in multinuclear osteoclasts and mononuclear phagocytes*



- In the marrow spaces, resorption causes a *thinning of the surrounding bony trabeculae and an enlargement of the marrow spaces---* destruction of bone and reduction in bone height



To Summarize...

- **Subgingival bacteria initiate and perpetuate the immune–inflammatory responses in the periodontal tissues--classic signs of inflammation**
- **Inflammatory responses to bacterial challenge are protective by intent, but they result in the majority of tissue damage and breakdown**
- **Individuals vary with regard to their susceptibility to periodontal disease which are genetically determined, and influenced by environmental risk factors (e.g., smoking)**
- **The challenge for the future is to identify at-risk individuals who possess the hyper-inflammatory trait so that disease can be prevented by careful management strategies before tissue loss has occurred**

References

Newman and Carranza's Clinical Periodontology and Implantology 14th Edition

Chapter 8: Periodontal Disease Pathogenesis Page no: 93-98

Chapter 10: Biofilm and Periodontal Microbiology Page no: 151-156



Thank you

